

동역학

(Engineering Mechanics: Dynamics)

Chapter 12-3

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Curvilinear Motion: Normal And Tangential Components

Determine the normal and tangential components of velocity and acceleration of a particle traveling along a curved path.



Cars traveling along a clover-leaf interchange experience an acceleration due to a change in velocity as well as due to a change in direction of the velocity.

If the car's speed is increasing at a known rate as it travels along a curve, how can we determine the magnitude and direction of its total acceleration?

Why would you care about the total acceleration of the car?

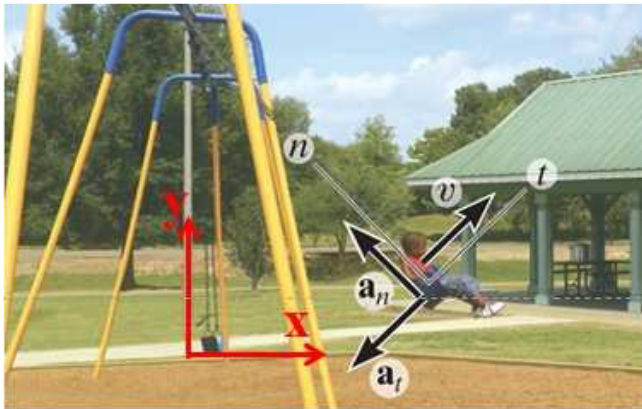


Applications

As the boy swings upward with a velocity $\mathbf{v}$, his motion can be analyzed using **n-t** coordinates.

As he rises, the magnitude of his velocity is changing, and thus his acceleration is also changing. How can we determine his velocity and acceleration at the bottom of the arc?

Can we use different coordinates, such as x-y coordinates, to describe his motion? Which coordinate system would be easier to use to describe his motion? Why?



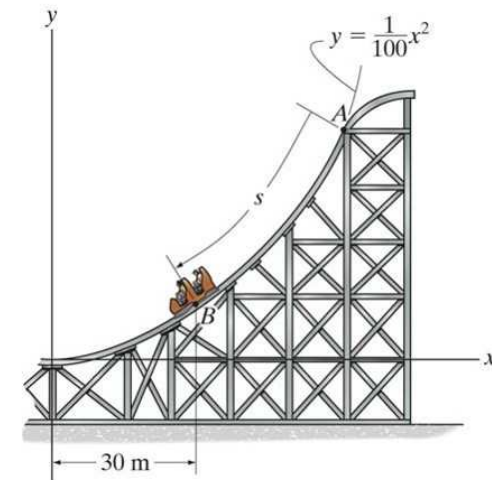
A roller coaster travels down a hill for which the path can be approximated by a function

$$y = f(x).$$

The roller coaster starts from rest and increases its speed at a constant rate.

How can we determine its velocity and acceleration at the bottom?

Why would we want to know these values?



Applications

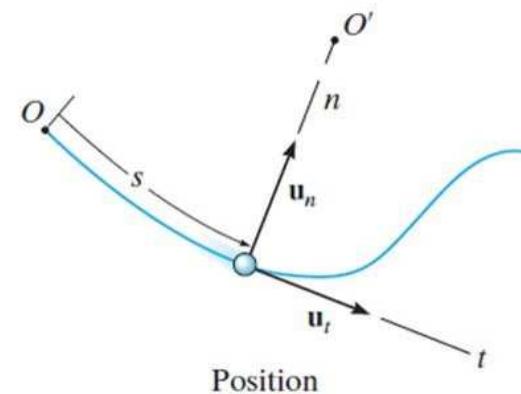
If we have an idea of the path of a vehicle or object, it is often convenient to analyze the motion using normal and tangential components (sometimes called *path* coordinates).



Normal and Tangential Components

When a particle moves along a curved path, it is sometimes convenient to describe its motion using coordinates other than Cartesian. When the path of motion is known, normal (n) and tangential (t) coordinates are often used.

In the n - t coordinate system, the origin is located on the particle (thus the origin and coordinate system move with the particle).

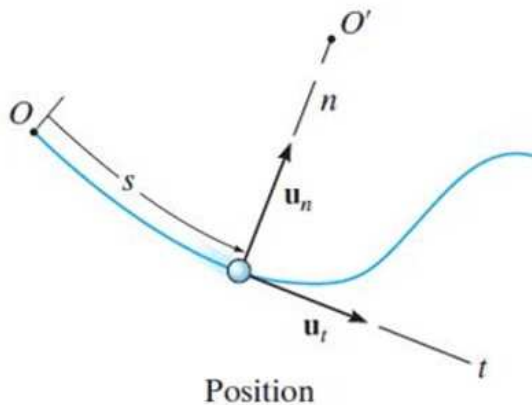


The t -axis is tangent to the path (curve) at the instant considered, positive in the direction of the particle's motion.

The n -axis is perpendicular to the t -axis with the positive direction toward the center of curvature of the curve.

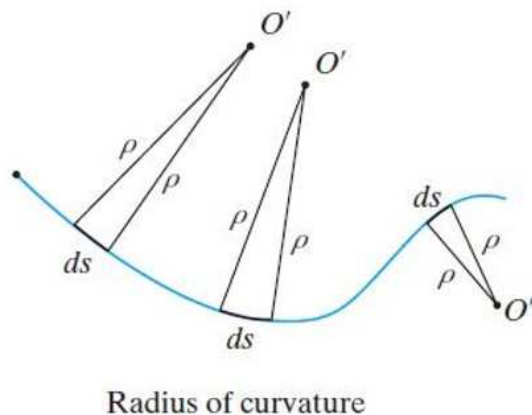


Normal and Tangential Components



The positive n and t directions are defined by the unit vectors u_n and u_t , respectively.

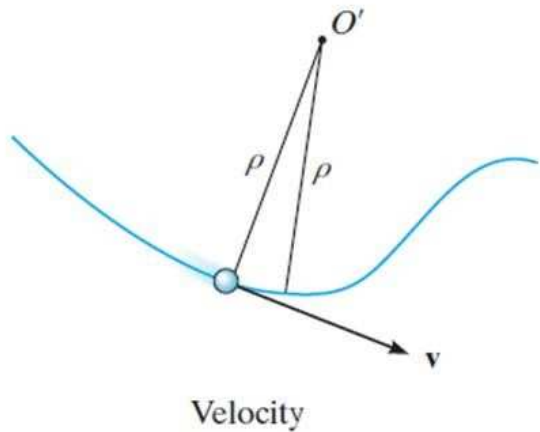
The center of curvature, O' , always lies on the concave side of the curve. The radius of curvature, ρ , is defined as the perpendicular distance from the curve to the center of curvature at that point.



The position of the particle at any instant is defined by the distance, s , along the curve from a fixed reference point.



Velocity in the **n-t** Coordinate System



The velocity vector is always tangent to the path of motion (t-direction).

The magnitude is determined by taking the time derivative of the path function, $s(t)$.

$$\mathbf{v} = v \mathbf{u}_t \quad \text{where} \quad v = \dot{s} = \frac{ds}{dt}$$

Here, v defines the magnitude of the velocity (speed) and $\mathbf{u}_t$ defines the direction of the velocity vector.



Acceleration in the **n-t** Coordinate System

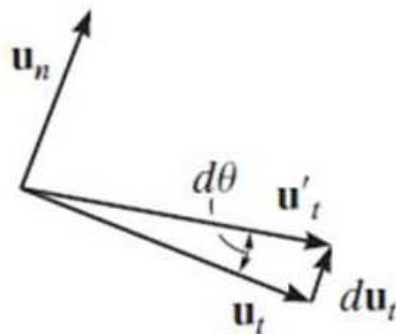
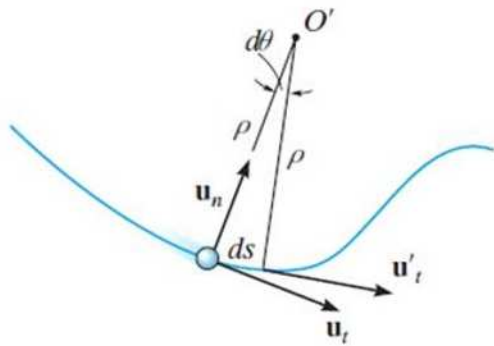
Acceleration is the time rate of change of velocity:

$$a = \frac{dv}{dt} = \frac{d(vu_t)}{dt} = \dot{v}u_t + v\dot{u}_t$$

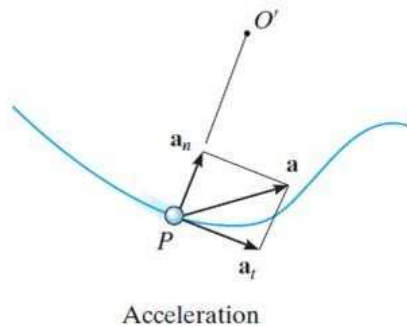
Here, $\dot{v}$ represents the change in the magnitude of velocity and, $\dot{u}_t$ represents the rate of change in the direction of u_t .

After mathematical manipulation, the acceleration vector can be expressed as:

$$a = \dot{v}u_t + \left(\frac{v^2}{\rho}\right)u_n = a_t u_t + a_n u_n.$$



Acceleration in the **n-t** Coordinate System



So, there are two components to the acceleration vector:

$$\mathbf{a} = a_t \mathbf{u}_t + a_n \mathbf{u}_n$$

- The tangential component is tangent to the curve and in the direction of increasing or decreasing velocity.

$$a_t = \dot{v} \quad \text{or} \quad a_t ds = v dv$$

- The normal or centripetal component is always directed toward the center of curvature of the curve.

$$a_n = \frac{v^2}{\rho}$$

- The magnitude of the acceleration vector is

$$a = \sqrt{(a_n)^2 + (a_t)^2}$$



Special Cases of Motion

There are some special cases of motion to consider.

1) The particle moves along a straight line.

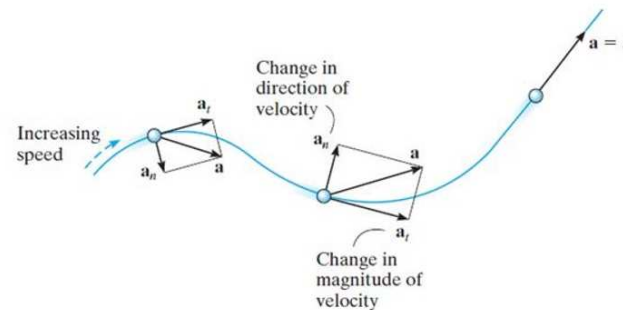
$$\rho \rightarrow \infty \Rightarrow a_n = \frac{v^2}{\rho} = 0 \Rightarrow a = a_t = \dot{v}$$

The tangential component represents the time rate of change in the magnitude of the velocity.

2) The particle moves along a curve at constant speed.

$$a_t = \dot{v} = 0 \Rightarrow a = a_n = \frac{v^2}{\rho}$$

The normal component represents the time rate of change in the direction of the velocity.



Special Cases of Motion

- 3) The tangential component of acceleration is constant, $a_t = (a_t)_c$.**

In this case,

$$s = s_o + v_o t + \left(\frac{1}{2}\right)(a_t)_c t^2$$

$$v = v_o + (a_t)_c t$$

$$v^2 = (v_o)^2 + 2(a_t)_c(s - s_o)$$

As before, s_o and v_o are the initial position and velocity of the particle at $t = 0$. How are these equations related to projectile motion equations? Why?

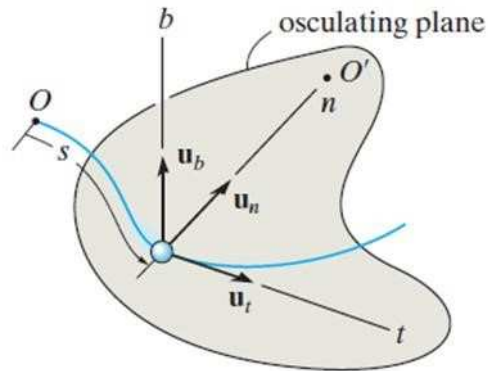
- 4) The particle moves along a path expressed as $y = f(x)$.**

The radius of curvature, ρ , at any point on the path can be calculated from

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}{\left|\frac{d^2y}{dx^2}\right|}$$



Three-dimensional Motion



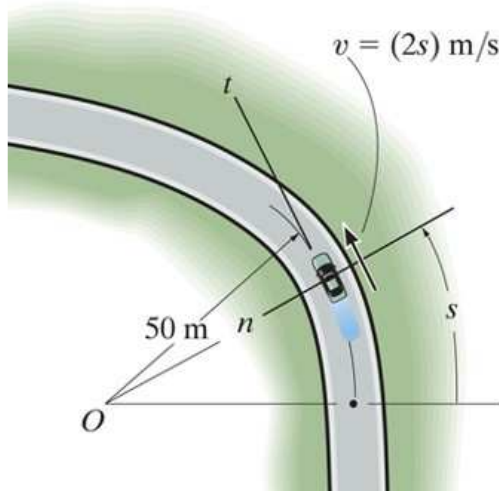
If a particle moves along a space curve, the n-t axes are defined as before. At any point, the t-axis is tangent to the path and the n-axis points toward the center of curvature. The plane containing the n-t axes is called the **osculating plane**.

A third axis can be defined, called the binomial axis, b. The binomial unit vector, u_b , is directed perpendicular to the osculating plane, and its sense is defined by the cross product $u_b = u_t \times u_n$.

There is no motion, thus no velocity or acceleration, in the binomial direction.



Example I: Car on the curved path



Given: A car travels along the road with a speed of $v = (2s) \text{ m/s}$, where s is in meters.

$$\rho = 50 \text{ m}$$

Find: The magnitudes of the car's acceleration at $s = 10 \text{ m}$.

- Plan:**
- 1) Calculate the velocity when $s = 10 \text{ m}$ using $v(s)$.
 - 2) Calculate the tangential and normal components of acceleration and then the magnitude of the acceleration vector.



Example I: Car on the curved path

Solutions:

- 1) The velocity vector is $\mathbf{v} = v\mathbf{u}_t$, where the magnitude is given by $v = (2s) \text{ m/s}$.

When $s = 10 \text{ m}$: $v = 20 \text{ m/s}$

- 2) The acceleration vector is $\mathbf{a} = a_t\mathbf{u}_t + a_n\mathbf{u}_n = \dot{v}\mathbf{u}_t + (v^2/\rho)\mathbf{u}_n$

Tangential component:

Since $a_t = \dot{v} = dv/dt = (dv/ds)(ds/dt) = v(dv/ds)$

where $v = 2s \Rightarrow a_t = d(2s)/ds (v) = 2v$

At $s = 10 \text{ m}$: $a_t = 40 \text{ m/s}^2$

Normal component: $a_n = v^2/\rho$

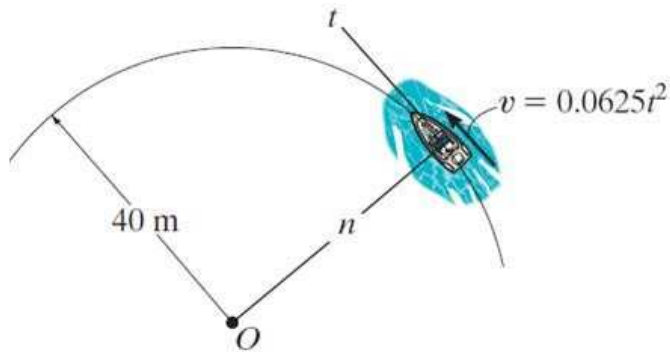
When $s = 10 \text{ m}$: $a_n = (20)^2/(50) = 8 \text{ m/s}^2$

The magnitude of the acceleration is

$$a = \sqrt{(a_n)^2 + (a_t)^2} = \sqrt{40^2 + 8^2} = 40.8 \text{ m/s}^2$$



Example II: Boat around a circular path



Given: A boat travels around a circular path, $\rho = 40$ m, at a speed that increases with time,

$$v = (0.0625 t^2) \text{ m/s.}$$

Find: The magnitudes of the boat's velocity and acceleration at the instant $t = 10$ s.

Plan: The boat starts from rest ($v = 0$ when $t = 0$).

- 1) Calculate the velocity at $t = 10$ s using $v(t)$.
- 2) Calculate the tangential and normal components of acceleration and then the magnitude of the acceleration vector.



Example II: Boat around a circular path

Solutions:

- 1) The velocity vector is $\mathbf{v} = v\mathbf{u}_t$, where the magnitude is given by

$$v = (0.0625t^2) \text{ m/s. At } t = 10\text{s:}$$

$$v = 0.0625 t^2 = 0.0625 (10)^2 = 6.25 \text{ m/s}$$

- 2) The acceleration vector is $\mathbf{a} = a_t\mathbf{u}_t + a_n\mathbf{u}_n = v\mathbf{u}_t + (v^2/\rho)\mathbf{u}_n$

Tangential component: $a_t = \dot{v} = d(0.0625 t^2)/dt = 0.125 t \text{ m/s}^2$

At $t = 10\text{s}$: $a_t = 0.125t = 0.125(10) = 1.25 \text{ m/s}^2$

Normal component: $a_n = v^2/\rho \text{ m/s}^2$

At $t = 10\text{s}$: $a_n = (6.25)^2 / (40) = 0.9766 \text{ m/s}^2$

The magnitude of the acceleration is

$$a = \sqrt{(a_n)^2 + (a_t)^2} = \sqrt{1.25^2 + 0.9766^2} = 1.59 \text{ m/s}^2$$



Concept Quiz

1. A particle traveling in a circular path of radius 300 m has an instantaneous velocity of 30 m/s and its velocity is increasing at a constant rate of 4 m/s^2 . What is the magnitude of its total acceleration at this instant?

A. 3 m/s^2

C. 5 m/s^2

B. 4 m/s^2

D. -5 m/s^2

2. If a particle moving in a circular path of radius 5 m has a velocity function $v = 4t^2 \text{ m/s}$, what is the magnitude of its total at $t = 1 \text{ s}$?

A. 8 m/s

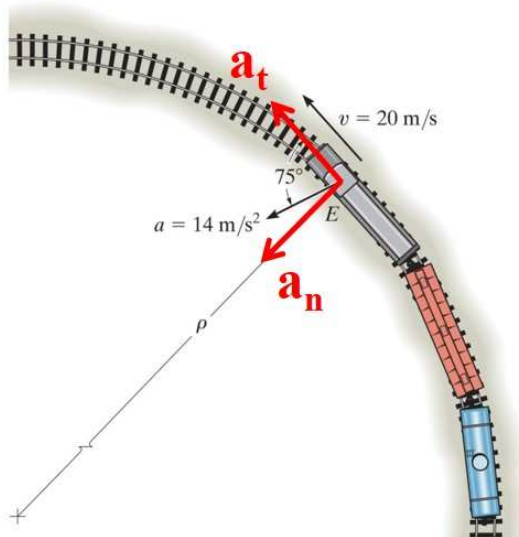
B. 8.6 m/s

C. 3.2 m/s

D. 11.2 m/s



Group Problem Solving I: Train Path



Given: The train engine at E has a speed of 20 m/s and an acceleration of 14 m/s^2 acting in the direction shown.

Find: The rate of increase in the train's speed and the radius of curvature ρ of the path.

- Plan:**
1. Determine the tangential and normal components of the acceleration.
 2. Calculate v from the tangential component of the acceleration.
 3. Calculate ρ from the normal component of the acceleration.



Group Problem Solving I: Train Path

Solutions:

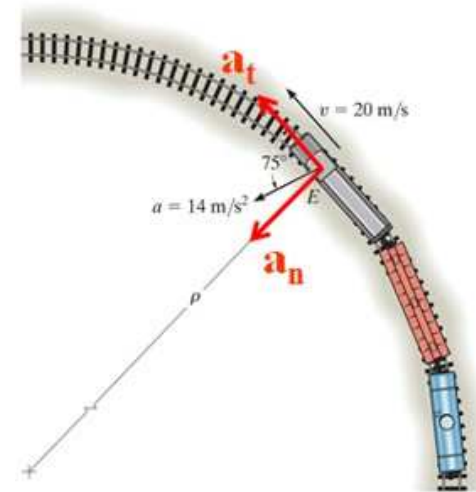
1) Acceleration

Tangential component:

$$a_t = 14 \cos(75) = 3.623 \text{ m/s}^2$$

Normal component:

$$a_n = 14 \sin(75) = 13.52 \text{ m/s}^2$$



2) The tangential component of acceleration is the rate of increase of the train's speed, so

$$a_t = \dot{v} = 3.62 \text{ m/s}^2.$$

3) The normal component of acceleration is

$$a_n = v^2 / \rho \Rightarrow 13.52 = 20^2 / \rho$$

$$\rho = 29.6 \text{ m}$$



Group Problem Solving II: Bicycle

Given: Starting from rest, a bicyclist travels around a horizontal circular path, $\rho = 10\text{m}$, at a speed of

$$v = (0.09 t^2 + 0.1 t) \frac{\text{m}}{\text{s}}.$$



Find: The magnitudes of her velocity and acceleration when she has travelled 3 m.

Plan: The bicyclist starts from rest ($v = 0$ when $t = 0$).

- 1) Integrate $v(t)$ to find the position $s(t)$.
- 2) Calculate the time when $s = 3\text{ m}$ using $s(t)$.
- 3) Calculate the tangential and normal components of acceleration and then the magnitude of the acceleration vector.



Group Problem Solving II: Bicycle

Solution:

- 1) The velocity vector is $v = (0.09 t^2 + 0.1 t) \frac{\text{m}}{\text{s}}$, where t is in seconds. Integrate the velocity and find the position $s(t)$.

$$\begin{aligned}\text{Position: } \int v dt &= \int (0.09 t^2 + 0.1 t) dt \\ s(t) &= 0.03 t^3 + 0.05 t^2\end{aligned}$$

- 2) Calculate the time, t when $s = 3$ m.

$$3 = 0.03 t^3 + 0.05 t^2$$

Solving for t , $t = 4.147$ s

The velocity at $t = 4.147$ s is,

$$v = 0.09 (4.147)^2 + 0.1(4.147) = 1.96 \text{ m/s}$$



Group Problem Solving II: Bicycle

3) The acceleration vector is $a = a_t u_t + a_n u_n = v u_t + (v^2/\rho) u_n$.

Tangential component:

$$a_t = \dot{v} = \frac{d(0.09t^2 + 0.1t)}{dt} = (0.18t + 0.1) \text{ m/s}^2$$

$$\text{At } t = 4.147 \text{ s : } a_t = 0.18(4.147) + 0.1 = 0.8465 \text{ m/s}^2$$

Normal component:

$$a_n = \frac{v^2}{\rho} \text{ m/s}^2$$

$$\text{At } t = 4.147 \text{ s : } a_n = \frac{(1.96)^2}{(10)} = 0.3852 \text{ m/s}^2$$

The magnitude of the acceleration is

$$a = \sqrt{(a_n)^2 + (a_t)^2} = \sqrt{0.8465^2 + 0.3852^2} = 0.930 \text{ m/s}^2$$



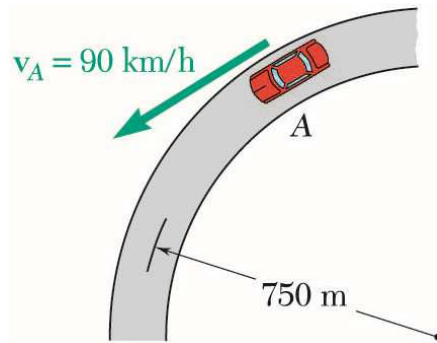
Attention Quiz

1. The magnitude of the normal acceleration is
 - A. proportional to radius of curvature.
 - B. inversely proportional to radius of curvature.
 - C. sometimes negative.
 - D. zero when velocity is constant.

2. The directions of the tangential acceleration and velocity are always
 - A. perpendicular to each other.
 - B. collinear.
 - C. in the same direction.
 - D. in opposite directions.



Sample Problem: Curved Highway



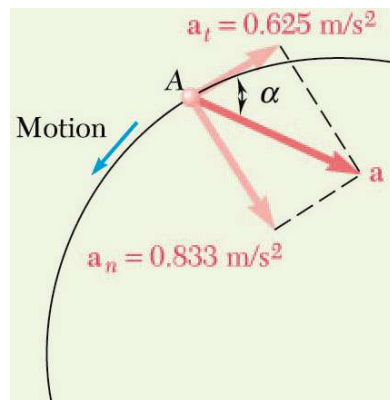
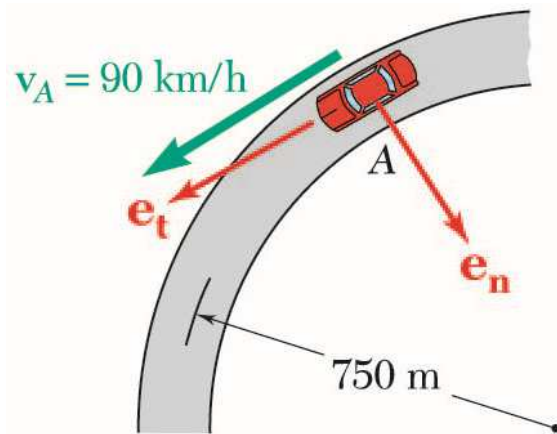
A motorist is traveling on a curved section of highway of radius 750 m at the speed of 90 km/h. The motorist suddenly applies the brakes, causing the automobile to slow down at a constant rate. Knowing that after 8 s the speed has been reduced to 72 km/h, determine the acceleration of the automobile immediately after the brakes have been applied.

STRATEGY:

- Define your coordinate system
- Calculate the tangential velocity and tangential acceleration
- Calculate the normal acceleration
- Determine overall acceleration magnitude after the brakes have been applied



Sample Problem: Curved Highway



MODELING and ANALYSIS:

- Define your coordinate system
- Determine velocity and acceleration in the tangential direction

$$90 \text{ km/h} = \left(90 \frac{\text{km}}{\text{h}}\right) \left(\frac{1000 \text{ m}}{1 \text{ km}}\right) \left(\frac{1 \text{ h}}{3600 \text{ s}}\right) = 25 \text{ m/s}$$

$$72 \text{ km/h} = 20 \text{ m/s}$$

- The deceleration constant, therefore

$$a_t = \text{average } a_t = \frac{\Delta v}{\Delta t} = \frac{20 \text{ m/s} - 25 \text{ m/s}}{8 \text{ s}} = -0.625 \text{ m/s}^2$$

- Immediately after the brakes are applied, the speed is still 25 m/s

$$a_n = \frac{v^2}{\rho} = \frac{(25 \text{ m/s})^2}{750 \text{ m}} = 0.833 \text{ m/s}^2$$

$$a = \sqrt{a_n^2 + a_t^2} = \sqrt{0.625^2 + 0.833^2} \quad \tan \alpha = \frac{a_n}{a_t} = \frac{0.833 \text{ m/s}^2}{0.625 \text{ m/s}^2}$$

$$a = 1.041 \text{ m/s}^2$$

$$\alpha = 53.1^\circ$$

More Applications

In 2001, a race scheduled at the Texas Motor Speedway was cancelled because the normal accelerations were too high and caused some drivers to experience excessive g-loads (similar to fighter pilots) and possibly pass out.

What are some things that could be done to solve this problem?

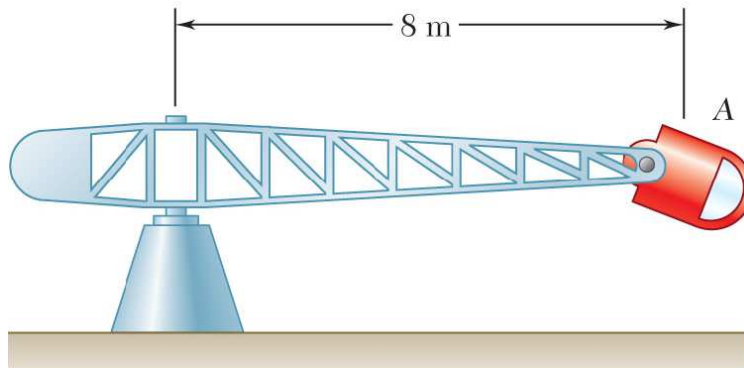


Some possibilities:

- Reduce the allowed speed
- Increase the turn radius (difficult and costly)
- Have the racers wear g-suits



Group Problem Solving: Centrifuge Cab



The tangential acceleration of the centrifuge cab is given by

$$a_t = 0.5t \text{ (m/s}^2\text{)}$$

where t is in seconds and a_t is in m/s^2 . If the centrifuge starts from rest, determine the total acceleration magnitude of the cab after 10 seconds.

STRATEGY:

- Define your coordinate system
- Calculate the tangential velocity and tangential acceleration
- Calculate the normal acceleration
- Determine overall acceleration magnitude



Group Problem Solving: Centrifuge Cab

MODELING and ANALYSIS:

Define your coordinate system

In the side view, the tangential direction points into the "page"

Determine the tangential velocity

$$a_t = 0.5t$$

$$v_t = \int_0^t 0.5t \, dt = 0.25t^2 \Big|_0^t = 0.25t^2$$

$$v_t = 0.25(10)^2 = 25 \text{ m/s}$$

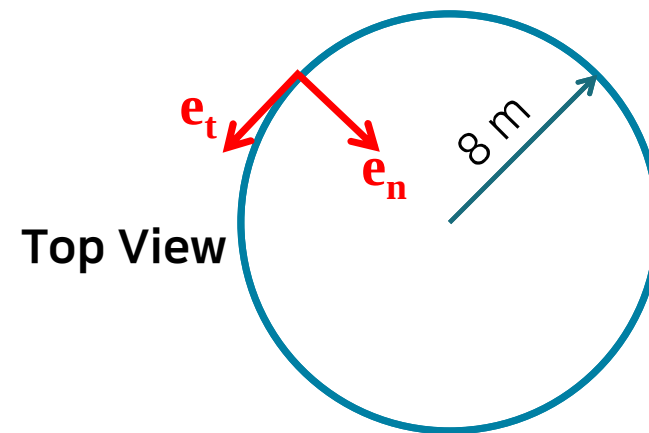
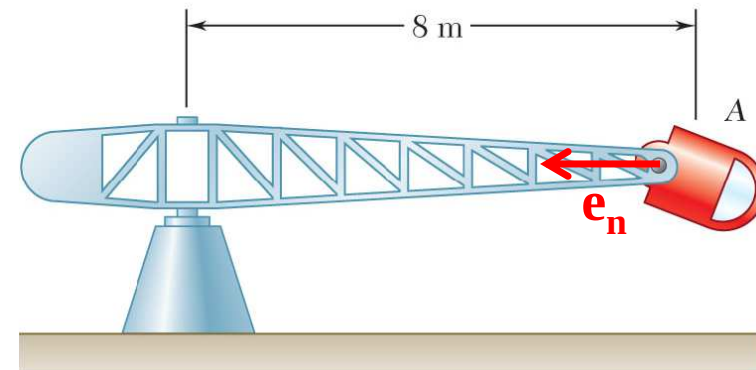
Determine the normal acceleration

$$a_n = \frac{(v_t)^2}{r} = \frac{25^2}{8} = 78.125 \text{ m/s}^2$$

Determine the total acceleration magnitude

$$a_{mag} = \sqrt{a_n^2 + a_t^2} = \sqrt{78.125^2 + [(0.5)(10)]^2}$$

$$a_{mag} = 78.3 \text{ m/s}^2$$



Curvilinear Motion: Cylindrical Components

(Radial and Transverse Components)

The foot pedal on an elliptical machine rotates about and extends from a central pivot point. This motion can be analyzed using cylindrical components

Fire truck ladders can rotate as well as extend; the motion of the end of the ladder can be analyzed using cylindrical components.

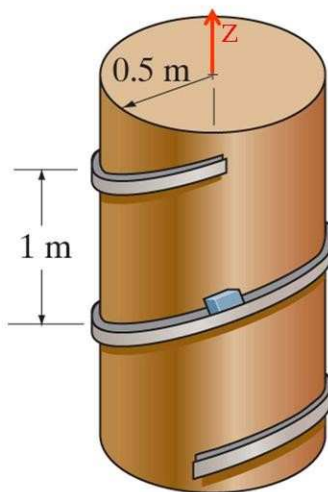
Determine velocity and acceleration components using **cylindrical coordinates.**



Applications

A cylindrical coordinate system is used in cases where the particle moves along a 3-D curve.

In the figure shown, the box slides down the helical ramp. How would you find the box's velocity components to check to see if the package will fly off the ramp?

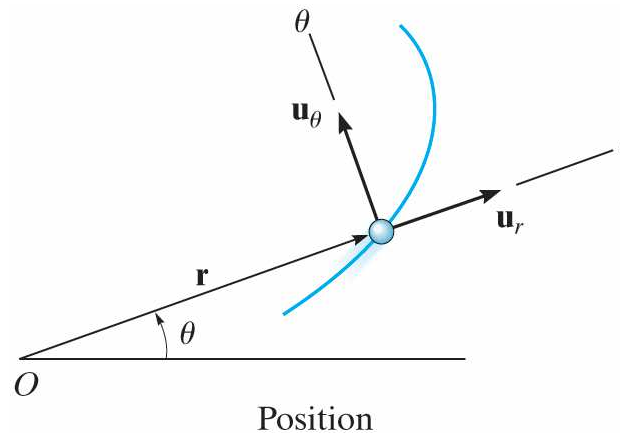


The cylindrical coordinate system can be used to describe the motion of the girl on the slide.

Here the **radial coordinate** is constant, the **transverse coordinate** increases with time as the girl rotates about the vertical axis, and her altitude, z , decreases with time. How can you find her acceleration components?



Cylindrical Components

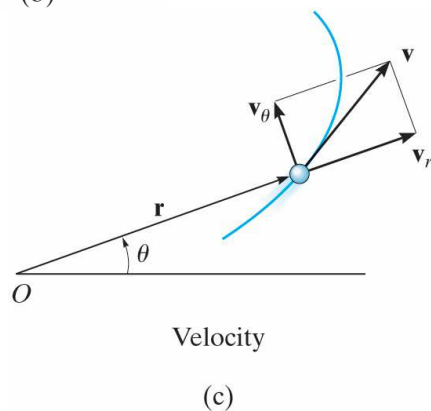
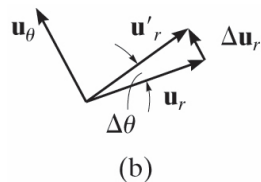
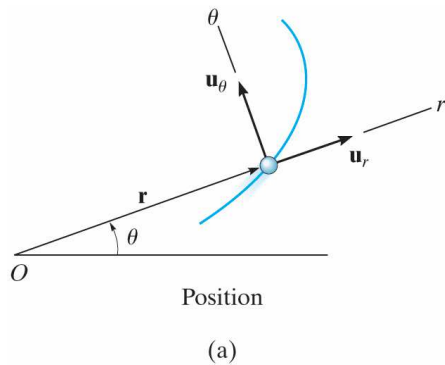


(a)

We can express the location of P in polar coordinates as $\mathbf{r} = r\mathbf{u}_r$. Note that the radial direction, r , extends outward from the fixed origin, O , and the transverse coordinate, θ is measured counter-clockwise (CCW) from the horizontal.



Velocity (Polar Coordinates)



The instantaneous velocity is defined as:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d(r\mathbf{u}_r)}{dt} \quad \mathbf{v} = \dot{r}\mathbf{u}_r + r\frac{d\mathbf{u}_r}{dt}$$

Using the chain rule: $\frac{d\mathbf{u}_r}{dt} = \left(\frac{d\mathbf{u}_r}{d\theta}\right)\left(\frac{d\theta}{dt}\right)$

We can prove that $\frac{d\mathbf{u}_r}{d\theta} = \mathbf{u}_\theta$ so $\frac{d\mathbf{u}_r}{dt} = \dot{\theta}\mathbf{u}_\theta$

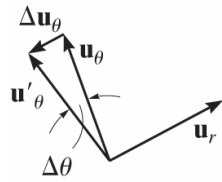
Therefore: $\mathbf{v} = \dot{r}\mathbf{u}_r + r\dot{\theta}\mathbf{u}_\theta$

Thus, the velocity vector has two components:

$\dot{r}$, called the radial component, and $r\dot{\theta}$ called the transverse component. The speed of the particle at any given instant is the sum of the squares of both components or

$$v = \sqrt{(r\dot{\theta})^2 + (\dot{r})^2}$$

Acceleration (Polar Coordinates)



(d)

The instantaneous acceleration is defined as:

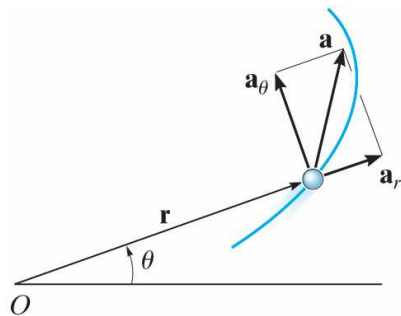
$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \left(\frac{d}{dt} \right) (\dot{r}\mathbf{u}_r + r\dot{\theta}\mathbf{u}_\theta)$$

After manipulation, the acceleration can be expressed as

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{u}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{u}_\theta$$

The term $(\ddot{r} - r\dot{\theta}^2)$ is the radial acceleration or a_r .

The term $(r\ddot{\theta} + 2\dot{r}\dot{\theta})$ is the transverse acceleration or a_θ .



Acceleration

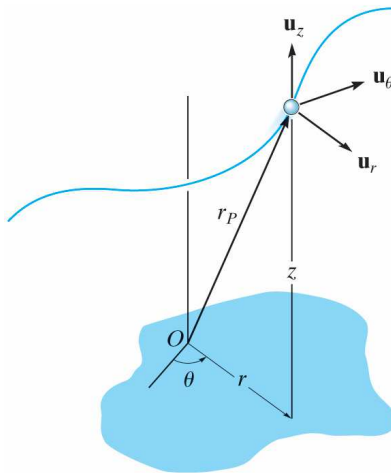
(e)

The magnitude of acceleration is

$$a = (\ddot{r} - r\dot{\theta}^2)^2 + (r\ddot{\theta} + 2\dot{r}\dot{\theta})^2$$



Cylindrical Coordinates



If the particle P moves along a space curve, its position can be written as

$$\mathbf{r}_P = r\mathbf{u}_r + z\mathbf{u}_z$$

Taking time derivatives and using the chain rule:

Velocity:

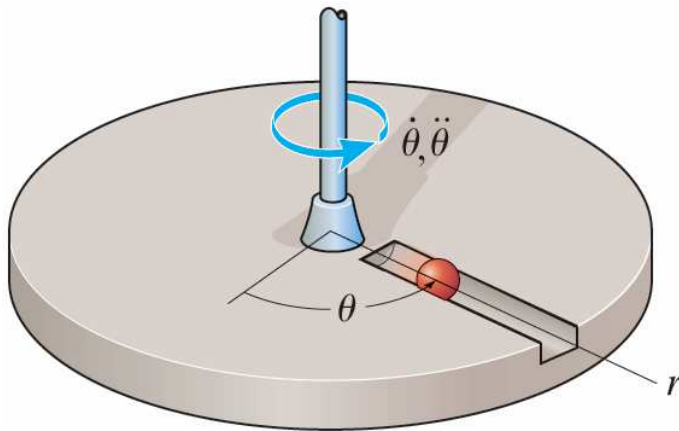
$$\mathbf{v}_P = \dot{r}\mathbf{u}_r + r\dot{\theta}\mathbf{u}_\theta + \dot{z}\mathbf{u}_z$$

Acceleration:

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{u}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{u}_\theta + \ddot{z}\mathbf{u}_z$$



Example: Rotating Platform



Given: The platform is rotating such that, at any instant, its angular position is $\theta = (4t^{3/2})$ rad, where t is in seconds.

A ball rolls outward so that its position is

$$r = (0.1t^3) \text{ m.}$$

Find: The magnitude of velocity and acceleration of the ball when $t = 1.5$ seconds.

Plan: Use a polar coordinate system and related kinematic equations.



Example: Rotating Platform

Solution:

$$r = 0.1t^3, \quad \dot{r} = 0.3t^2 \quad \ddot{r} = 0.6t$$

$$\theta = 4t^{3/2}, \quad \dot{\theta} = 6t^{1/2}, \quad \ddot{\theta} = 3t^{-1/2}$$

At $t=1.5$ s,

$$r = 0.3375 \text{ m}, \quad \dot{r} = 0.675 \text{ m/s}, \quad \ddot{r} = 2.449 \text{ m/s}^2$$

$$\theta = 7.348 \text{ rad}, \quad \dot{\theta} = 7.348 \text{ rad/s}, \quad \ddot{\theta} = 2.449 \text{ rad/s}^2$$

Substitute into the equation for velocity

$$\begin{aligned} v &= \dot{r}u_r + r\dot{\theta}u_\theta = 0.675u_r + 0.3375(7.348)u_\theta \\ &= 0.675u_r + 2.480u_\theta \end{aligned}$$

$$v = \sqrt{(0.675)^2 + (2.480)^2} = 2.57 \text{ m/s}$$



Example: Rotating Platform

Since car **B** is traveling along a curve, the acceleration of **B** is:

$$a = (\ddot{r} - r\dot{\theta}^2)u_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})u_\theta$$

$$a = [0.9 - 0.3375(7.348)^2]u_r \\ + [0.3375(2.449) + 2(0.675)(7.348)]u_\theta$$

$$a = -17.33 u_r + 10.75 u_\theta \text{ m/s}^2$$

$$a = \sqrt{(-17.33)^2 + (10.75)^2} = 20.4 \text{ m/s}^2$$



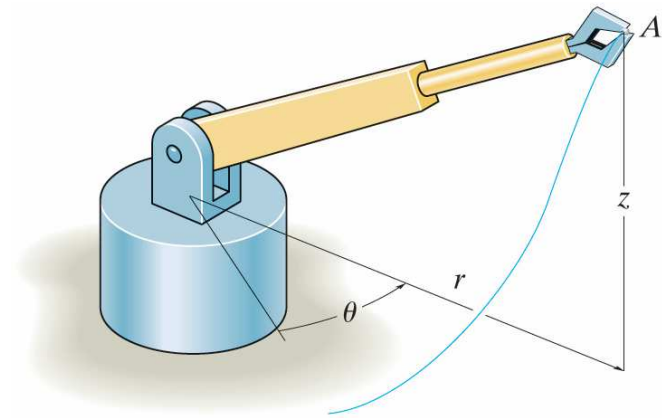
Concept Quiz

1. If $\dot{r}$ is zero for a particle, the particle is
 - A. not moving.
 - B. moving in a circular path.
 - C. moving on a straight line.
 - D. moving with constant velocity.

2. If a particle moves in a circular path with constant velocity, its radial acceleration is
 - A. zero.
 - B. $\ddot{r}$.
 - C. $-r\dot{\theta}^2$.
 - D. $2\dot{r}\dot{\theta}$.



Group Problem Solving I: Robot Arm



Given: The arm of the robot is extending at a constant rate

$$\dot{r} = 1.5 \text{ m/s when } r = 3 \text{ m,}$$

$$z = (4t^2) \text{ m, and } \theta = (0.5 t) \text{ rad,}$$

where t is in seconds.

Find: The velocity and acceleration of the grip A when $t = 3 \text{ s}$.

Plan: Use cylindrical coordinates.



Group Problem Solving I: Robot Arm

Solution:

When $t = 3 \text{ s}$, $r = 3 \text{ m}$ and the arm is extending at a constant rate

$$\dot{r} = 1.5 \text{ m/s. Thus } \ddot{r} = 0 \text{ m/s}^2$$

$$\theta = 0.5 t = 4.5 \text{ rad, } \dot{\theta} = 0.5 \text{ rad/s, } \ddot{\theta} = 0 \text{ rad/s}^2$$

$$z = 4 t^2 = 36 \text{ m, } \dot{z} = 8 t = 24 \text{ m/s, } \ddot{z} = 8 \text{ m/s}^2$$

Substitute in the equation for velocity

$$\begin{aligned} \mathbf{v} &= \dot{r}\mathbf{u}_r + r\dot{\theta}\mathbf{u}_\theta + \dot{z}\mathbf{u}_z \\ &= 1.5\mathbf{u}_r + 3(0.5)\mathbf{u}_\theta + 24\mathbf{u}_z \\ &= 1.5\mathbf{u}_r + 1.5\mathbf{u}_\theta + 24\mathbf{u}_z \end{aligned}$$

Magnitude

$$v = \sqrt{(1.5)^2 + (1.5)^2 + (24)^2} = 24.1 \text{ m/s}$$



Group Problem Solving I: Robot Arm

Acceleration equation in cylindrical coordinates

$$\begin{aligned} a &= (\ddot{r} - r\dot{\theta}^2)u_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})u_\theta + \ddot{z}u_z \\ &= \{0 - 3(0.5)^2\}u_r + \{3(0) + 2(1.5)0.5\}u_\theta + 8u_z \end{aligned}$$

$$a = [-0.75 u_r + 1.5 u_\theta + 8 u_z] \text{ m/s}^2$$

$$a = \sqrt{(-0.75)^2 + (1.5)^2 + (8)^2} = 8.17 \text{ m/s}^2$$



Attention Quiz

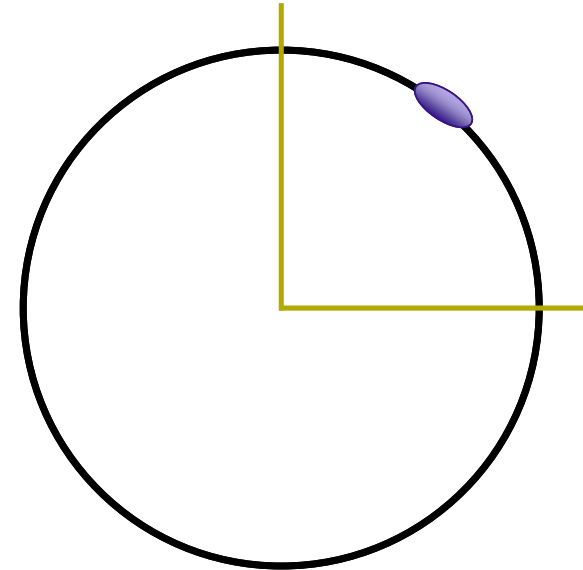
1. The radial component of velocity of a particle moving in a circular path is always
 - A. zero.
 - B. constant.
 - C. greater than its transverse component.
 - D. less than its transverse component.

2. The radial component of acceleration of a particle moving in a circular path is always
 - A. negative.
 - B. directed toward the center of the path.
 - C. perpendicular to the transverse component of acceleration.
 - D. All of the above.



Concept Quiz

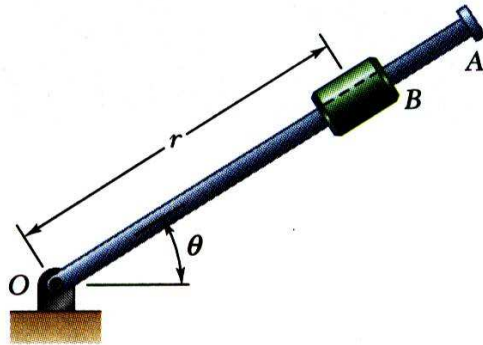
If you are travelling in a perfect circle, what is always true about radial/transverse coordinates and normal/tangential coordinates?



- a) The e_r direction is identical to the e_n direction.
- ☒ b) The e_θ direction is perpendicular to the e_n direction.
- c) The e_θ direction is parallel to the e_r direction.



Sample Problem: Rotating Arm and Collar



Rotation of the arm about O is defined by $\theta = 0.15t^2$ where θ is in radians and t in seconds. Collar B slides along the arm such that $r = 0.9 - 0.12t^2$ where r is in meters.

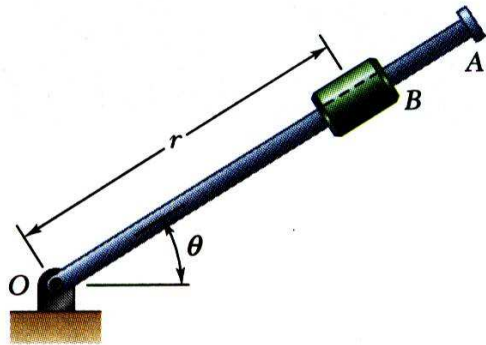
After the arm has rotated through 30° , determine (a) the total velocity of the collar, (b) the total acceleration of the collar, and (c) the relative acceleration of the collar with respect to the arm.

STRATEGY:

- Evaluate time t for $\theta = 30^\circ$.
- Evaluate radial and angular positions, and first and second derivatives at time t .
- Calculate velocity and acceleration in cylindrical coordinates.
- Evaluate acceleration with respect to arm.



Sample Problem: Rotating Arm and Collar



MODELING and ANALYSIS

- Evaluate time t for $\theta = 30^\circ$.

$$\begin{aligned}\theta &= 0.15t^2 \\ &= 30^\circ = 0.524\text{rad} \quad t = 1.869\text{ s}\end{aligned}$$

- Evaluate radial and angular positions, and first and second derivatives at time t .

$$r = 0.9 - 0.12t^2 = 0.481\text{ m}$$

$$\dot{r} = -0.24t = -0.449\text{ m/s}$$

$$\ddot{r} = -0.24\text{ m/s}^2$$

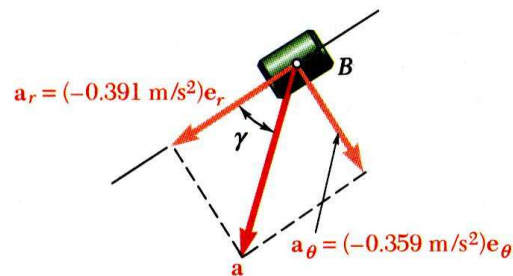
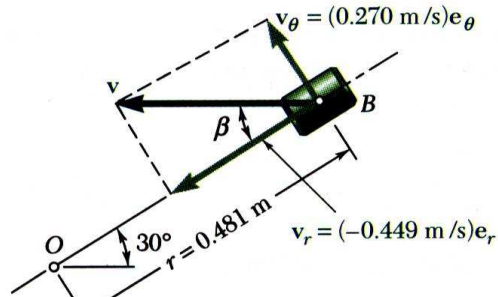
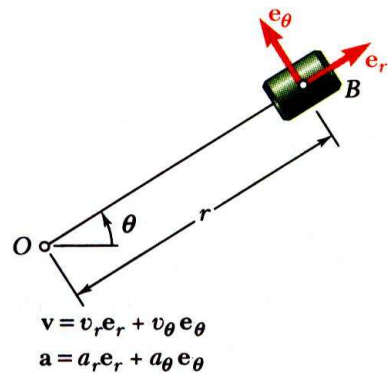
$$\theta = 0.15t^2 = 0.524\text{rad}$$

$$\dot{\theta} = 0.30t = 0.561\text{ rad/s}$$

$$\ddot{\theta} = 0.30\text{ rad/s}^2$$



Sample Problem: Rotating Arm and Collar



- Calculate velocity and acceleration.

$$v_r = \dot{r} = -0.449 \text{ m/s}$$

$$v_\theta = r\dot{\theta} = (0.481 \text{ m})(0.561 \text{ rad/s}) = 0.270 \text{ m/s}$$

$$v = \sqrt{v_r^2 + v_\theta^2} \quad \beta = \tan^{-1} \frac{v_\theta}{v_r}$$

$$v = 0.524 \text{ m/s} \quad \beta = 31.0^\circ$$

$$a_r = \ddot{r} - r\dot{\theta}^2$$

$$= -0.240 \text{ m/s}^2 - (0.481 \text{ m})(0.561 \text{ rad/s})^2$$

$$= -0.391 \text{ m/s}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

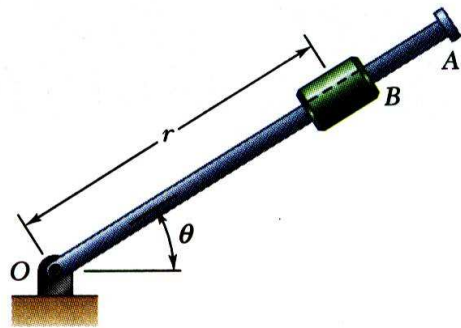
$$= (0.481 \text{ m})(0.3 \text{ rad/s}^2) + 2(-0.449 \text{ m/s})(0.561 \text{ rad/s})$$

$$= -0.359 \text{ m/s}^2$$

$$a = \sqrt{a_r^2 + a_\theta^2} \quad \gamma = \tan^{-1} \frac{a_\theta}{a_r}$$

$$a = 0.531 \text{ m/s}^2 \quad \gamma = 42.6^\circ$$

Sample Problem: Rotating Arm and Collar

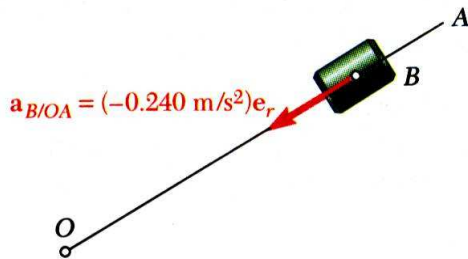


- Evaluate acceleration with respect to arm.
- Motion of collar with respect to arm is rectilinear and defined by coordinate r .

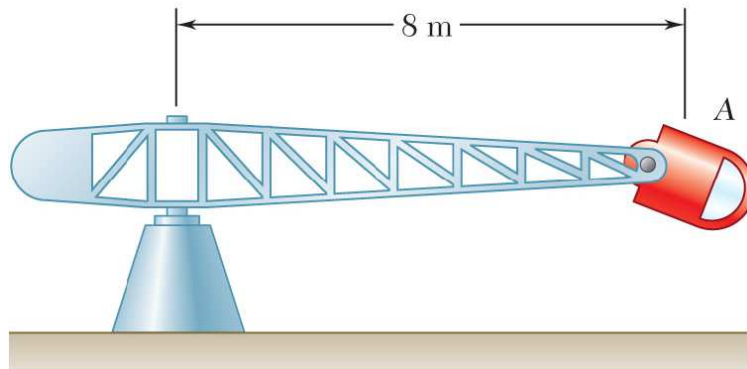
$$a_{B/OA} = \ddot{r} = -0.240 \text{ m/s}^2$$

REFLECT and THINK:

You should consider polar coordinates for any kind of rotational motion. They turn this problem into a straightforward solution, whereas any other coordinate system would make this problem much more difficult. One way to make this problem harder would be to ask you to find the radius of curvature in addition to the velocity and acceleration. To do this, you would have to find the normal component of the acceleration; that is, the component of acceleration that is perpendicular to the tangential direction defined by the velocity vector.



Group Problem Solving: Centrifuge Arm



The angular acceleration of the centrifuge arm varies according to

$$\ddot{\theta} = 0.05\theta \text{ (rad/s}^2\text{)}$$

where θ is measured in radians. If the centrifuge starts from rest, determine the acceleration magnitude after the gondola has travelled two full rotations.

STRATEGY:

- Define your coordinate system
- Calculate the angular velocity after three revolutions
- Calculate the radial and transverse accelerations
- Determine overall acceleration magnitude



Group Problem Solving: Centrifuge Arm

MODELING and ANALYSIS:

Define your coordinate system

In the side view, the transverse direction points into the "page"

Determine the angular velocity

$$\ddot{\theta} = 0.05\theta \text{ (rad/s}^2\text{)}$$

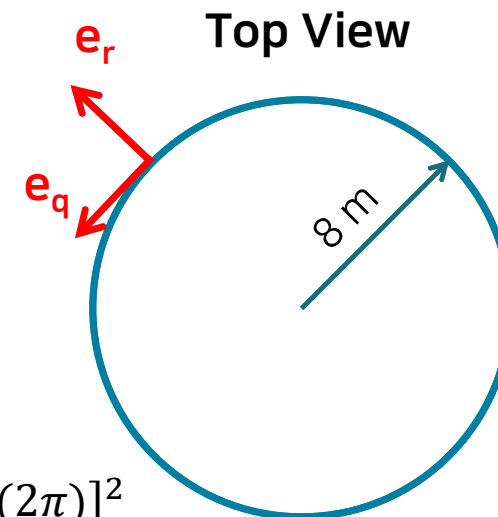
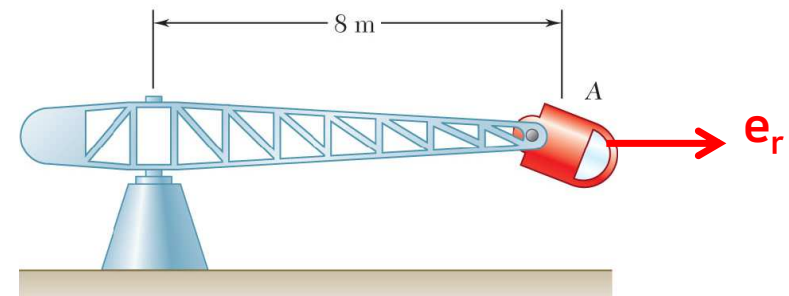
Acceleration is a function of position, so use: $\ddot{\theta}d\theta = \dot{\theta}d\dot{\theta}$

Evaluate the integral

$$\int_0^{(2)(2\pi)} 0.05\theta d\theta = \int_0^{\dot{\theta}} \dot{\theta} d\dot{\theta}$$

$$\frac{0.05\theta^2}{2} \Big|_0^{2(2\pi)} = \frac{\dot{\theta}^2}{2} \Big|_0^{\dot{\theta}}$$

$$\dot{\theta}^2 = 0.05[2(2\pi)]^2$$

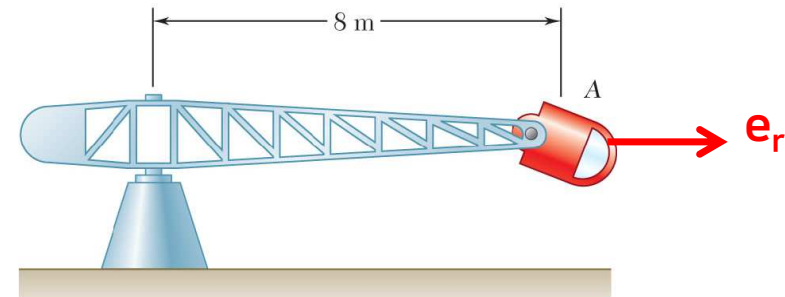


Group Problem Solving: Centrifuge Arm

Determine the angular velocity

$$\dot{\theta}^2 = 0.05[2(2\pi)]^2$$

$$\dot{\theta} = 2.8099 \text{ rad/s}$$



Determine the angular acceleration

$$\ddot{\theta} = 0.05\dot{\theta} = 0.05(2)(2\pi) = 0.6283 \text{ rad/s}^2$$

Find the radial and transverse accelerations

$$\begin{aligned}\vec{a} &= (\ddot{r} - r\dot{\theta}^2)\vec{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\vec{e}_\theta \\ &= (0 - (8)(2.8099)^2)\vec{e}_r + ((8)(0.6283) + 0)\vec{e}_\theta \\ &= -63.166\vec{e}_r + 5.0265\vec{e}_\theta \text{ (m/s}^2\text{)}\end{aligned}$$

Magnitude:

$$a_{mag} = \sqrt{a_r^2 + a_\theta^2} = \sqrt{(-63.166)^2 + [5.0265]^2}$$

$$a_{mag} = 63.4 \text{ m/s}^2$$

